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Informatica e Laboratorio di Programmazione

The true sign of intelligence is not knowledge but imagination.

Albert Einstein

Lessons

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- ~ ⅔ in Class (slides + code examples)
- ~ ⅓ in Laboratory (actual programming)
- Slides, code examples, and laboratory assignments
  - https://elly2024.didattica.unipr.it
  - https://www.ce.unipr.it/didattica/informatica/
  - Warning: they are continuously updated!

Schedule

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- September 19th – December 18th
  - Monday 13:30-15:30 (Lab. E c/o Q02)
  - Wednesday 10:30-12:30 (Aula F c/o sede didattica ing.)
  - Thursday 10:30-12:30 (Aula F c/o sede didattica ing.)

Calendar

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- Pauses
  - November 18th → Elections?

Prerequisites

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- Candidates should
  - know basic concepts of Information Technology
  - be able to use PCs and deal with files
  - know how to read and write....

Course program

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- Theory
  - Class
- Practical part AKA programming
  - Class + Laboratory
- Problem solving
  - Class + Laboratory

Course organization

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- Theory (Class)
- Basic information needed to understand how to program
  - Basic computer architecture
  - Algorithms
  - How numbers can be managed in a computer

Course organization

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- Programming part (class + laboratory)
  - Centered on the programming language and algorithms
  - C fundamentals
  - Variables
  - Control instructions
  - Expressions
  - Array, strings & pointers
  - Functions and recursion
  - I/O
- Slides + code examples

Course organization

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- How to access laboratories
- You **must** use your official UNIPR account:
  - name.surnameXY@studenti.unipr.it
- Temporary you can use “exam” accounts:
  - User: workstation name, Password: “noncopiare”
- Feel free to use your laptop/notebook
  - Caveat: during the exam you can not do that!

Course organization

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- Problem solving (class + laboratory)
- Laboratory assignements and old exam for practicing
  - live

Exam Rules

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- C programming in laboratory
  - Initially we will discuss the assignment all together
  - ~2 hours
  - C
  - Modular approach
- Rating
  - No errors (and also warnings) when compiling
  - It must be executable
  - Please, try to pass the exam when you are ready

Exam results

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- https://www.ce.unipr.it/didattica/informatica/esami/

UNIVERSITÀ DI PARMA Correzioni online di Informatica & Laboratorio di Programmazione

Menù

Da queste pagine potete accedere alle correzioni online delle prove pratiche di Informatica & Laboratorio di Programmazione
Ritorna alla pagina del corso.

Elenco prove pratiche:

A.A. 2023-2024

1. 18 settembre 2024 *Allocazione dinamica di array bidimensionali, battaglia navale*
2. 2 settembre 2024 *Memorizzazione dinamica di coppie di stringhe, palindromi e anagrammi*
3. 31 luglio 2024 *Problema di Milena, lettura da file ASCII con righe con lunghezze differenti, calcolo massimo*
4. 17 giugno 2024 *Lettura, memorizzazione e somma polinomi. Array bidimensionali di numeri int*
5. 4 aprile 2024 *Allocazione dinamica di un vettore, calcolo numero divisori primi*
6. 19 febbraio 2024 *Controllo validità quadrati magici, allocazione dinamica matrici*
7. 15 gennaio 2024 *Lettura file CSV in array di struct, ordinamento con qsort()*
8. 21 dicembre 2023 *Lettura di file contenenti cruciverba, array bidimensionali di char, conversione di array di char in stringhe*

A.A. 2022-2023

1. 15 settembre 2023 *Lettura file con array di stringhe, creazione striga di codifica*
2. 4 settembre 2023 *Problema di Giuseppe, analisi array e relative allocazioni*

How to approach this course

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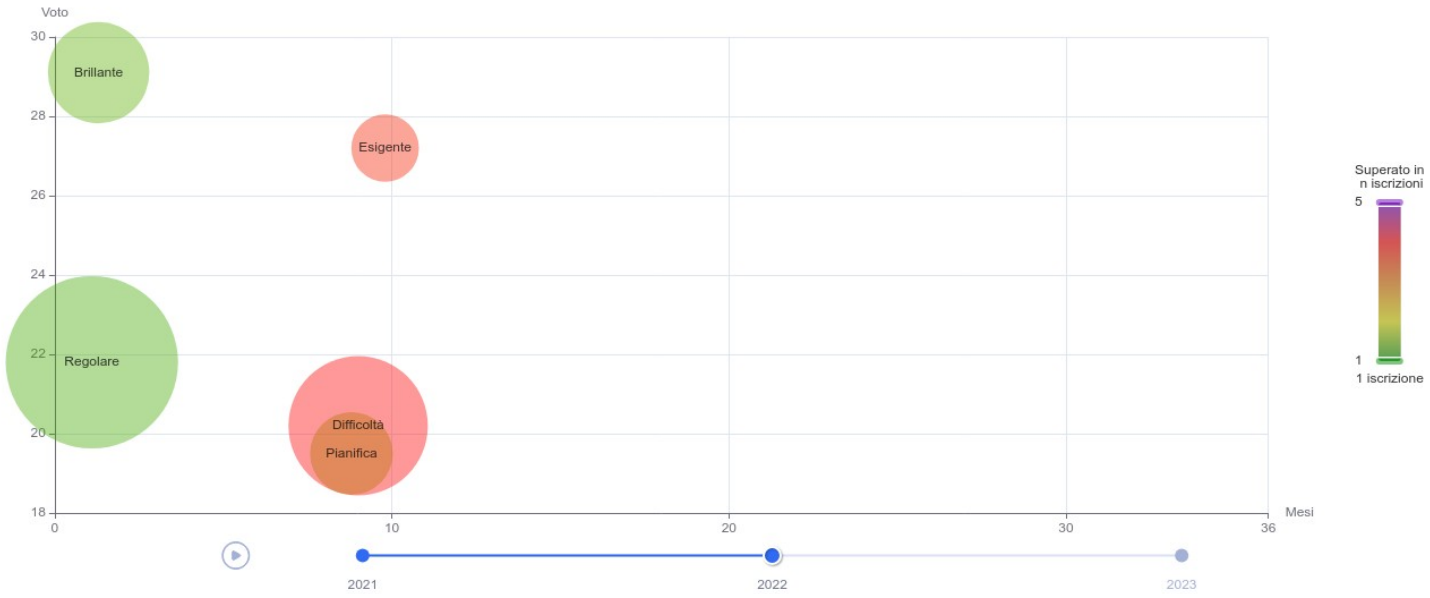
- Do not underestimate the effort needed to pass the exam
- Usually students that do not attend lessons ⟶ negative result
- Some statistics (2022 cohort)
  - Passed: 56.5% (91 students out of 161)
  - Not Passed: 43.5% (70 students out of 161)
  - Time average : 1.7 months
  - Average n. of attempts: 1.9
  - Average result: 23.5

How to approach this course



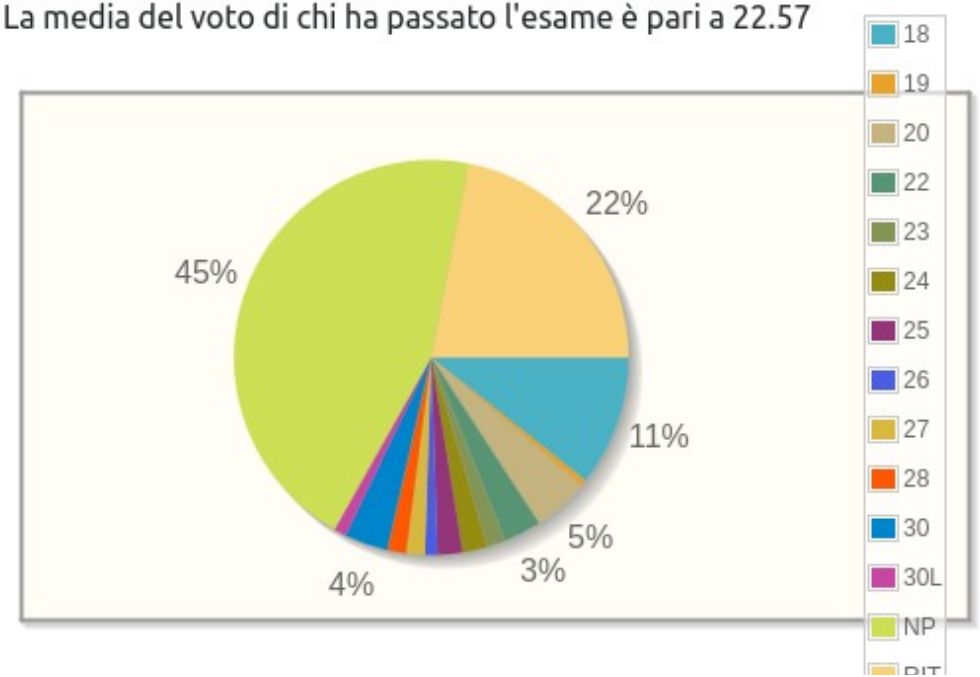
Regular → less than 3 attempts, before 6 months from 1st registration, and result < 27
Brilliant → less than 3 attempts, before 6 months from 1st registration, and result ≥ 27
Demanding → more than 3 registrations ≥ 27
Difficult → more than 3 registrations < 27
Planned → less than 3 registrations but after 6 months

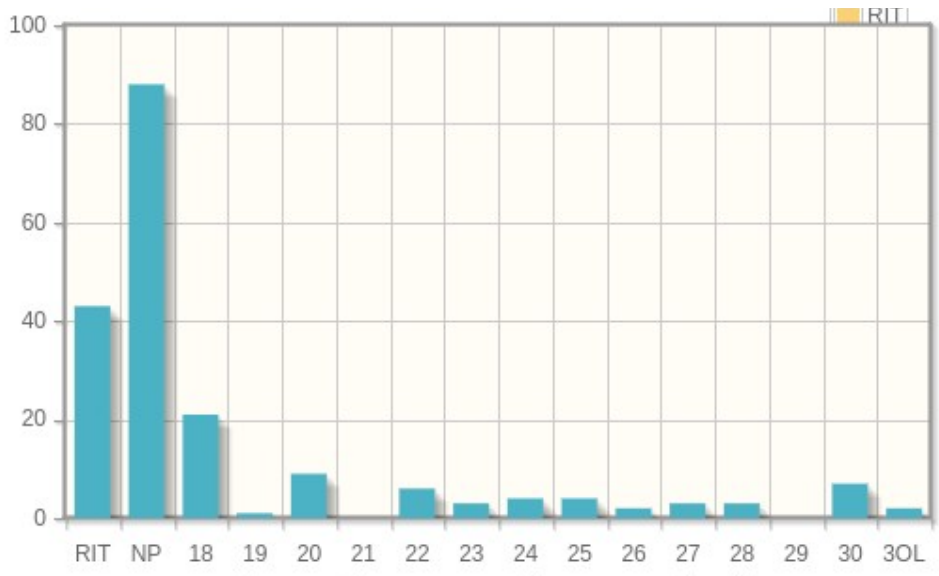
How to approach this course

Statistiche dei risultati dall'A.A. 2023-2024

Hanno passato la prova: 65 su 153 con un tasso pari a 42.48% oppure pari a 33.16% considerando anche i ritirati

La media del voto di chi ha passato l'esame è pari a 22.57





Student reception

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- Tuesday afternoon (tentatively) > 16:00
- Bldg. 1 of Cubes (scientific venue of engineering), 1st floor

Course material

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- You can download
  - Slides we discuss during lessons
  - C examples we use with slides
  - Assignments we use in laboratory + code
- Slides & Code examples are mostly definitive (besides small corrections)
- Assignments ⟶ continuously updated

Course material

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- How to download it?
  - Github!
  - Via web
  - Using “git” o “repo”
- Detailed instructions at:
  - https://www.ce.unipr.it/didattica/informatica/repo.php

Textbooks

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- Good C manual
- Suggestions:
  - K.N. King, *C Programming: A Modern Approach (2nd Edition)*, W W Norton & Co
  - Bellini Guidi, *Linguaggio C*, Mc Graw Hill
  - Darnell Margolis, *C manuale di programmazione*, Mc Graw Hill

Returns c (the character written). If a write error occurs, `putchar` sets the stream's error indicator and returns `EOF`. *7.3, 22.4*

`puts` *Write String* `<stdio.h>`

```
int puts(const char *s);
```

Writes the string pointed to by `s` to the `stdout` stream, then writes a new-line character.

Returns A nonnegative value if successful. Returns `EOF` if a write error occurs. *13.3, 22.5*

`putwc` *Write Wide Character to File (C99)* `<wchar.h>`

```
wint_t putwc(wchar_t c, FILE *stream);
```

Wide-character version of `putc`. *25.5*

`putwchar` *Write Wide Character (C99)* `<wchar.h>`

```
wint_t putwchar(wchar_t c);
```

Wide-character version of `putchar`. *25.5*

`qsort` *Sort Array* `<stdlib.h>`

```
void qsort(void *base, size_t nmemb, size_t size,
           int (*compar)(const void *, const void *));
```

Sorts the array pointed to by `base`. The array has `nmemb` elements, each `size` bytes long. `compar` is a pointer to a comparison function. When passed pointers to two array elements, the comparison function must return a negative, zero, or positive integer, depending on whether the first array element is less than, equal to, or greater than the second. *17.7, 26.2*

`raise` *Raise Signal* `<signal.h>`

```
int raise(int sig);
```

Raises the signal whose number is `sig`.

Returns Zero if successful, nonzero otherwise. *24.3*

`rand` *Generate Pseudo-Random Number* `<stdlib.h>`

```
int rand(void);
```

Returns A pseudo-random integer between 0 and `RAND_MAX` (inclusive). *26.2*

`realloc` *Resize Memory Block* `<stdlib.h>`

```
void *realloc(void *ptr, size_t size);
```

`ptr` is assumed to point to a block of memory previously obtained from `calloc`, `malloc`, or `realloc`. `realloc` allocates a block of `size` bytes, copying the contents of the old block if necessary.

Returns A pointer to the beginning of the new memory block. Returns a null pointer if a block of the requested size can't be allocated. *17.3*

A.14.2 Pseudo-Random Number Generator Functions

The *rand()* and *srand()* functions enable you to generate pseudo-random numbers.

A.14.2.1 The *rand()* Function

```
#include <stdlib.h>
int rand( void );
```

The *rand()* function returns an integer in the range 0 through RAND_MAX. Successive calls to *rand()* should produce different integers. However, the sequence of random numbers could be the same for each program execution unless you use a different seed value via the *srand()* function.

A.14.2.2 The *srand()* Function

```
#include <stdlib.h>
void srand( unsigned int seed );
```

The *srand()* function uses the argument as a seed for a new sequence of pseudo-random numbers to be returned by subsequent calls to *rand()*. If *srand()* is invoked with the same seed value, the sequence of generated numbers will be the same. The default seed value is 1.

A.14.3 Memory Management Functions

The memory management functions enable you to allocate and deallocate memory dynamically. See Chapter 7 for more information about these functions.

A.14.3.1 The *calloc()* Function

```
#include <stdlib.h>
void *calloc( size_t nmemb, size_t size );
```

The *calloc()* function allocates contiguous space for *nmemb* objects, each of which has a length in bytes specified by *size*. All bits in the allocated space are initialized to zero. *calloc()* returns a pointer to the first byte of the allocated space. If the space cannot be allocated, or if *nelem* or *size* is zero, *calloc()* returns a null pointer.

Contacts

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nec scire fas est omnia

Quinto Orazio Flacco, Odi